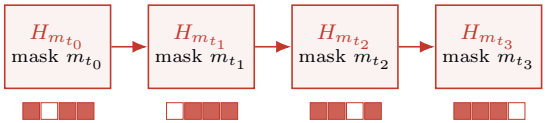
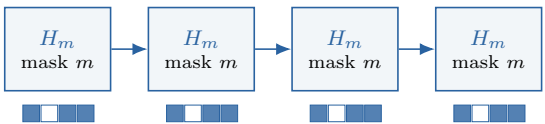
(a) Mechanism of trajectory-consistent dropout

Standard fresh-mask MC: mask resampled at every leapfrog sub-step ⇒ H_{m_t} varies in t



no single H integrated

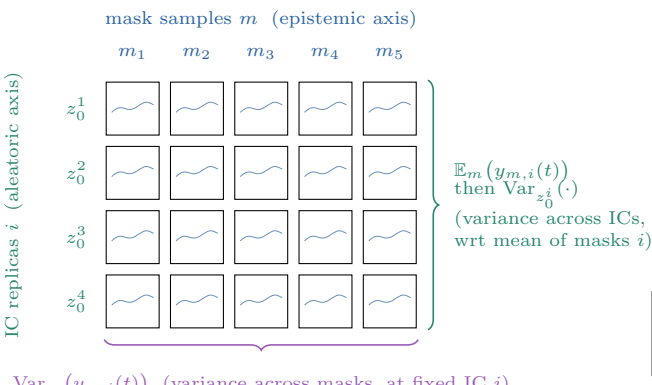
Fixed-mask MC (ours): one mask m sampled per rollout, reused across all sub-steps ⇒ coherent H_m



single sampled Hamiltonian H_m

leapfrog sub-step t

(b) IC ensemble separates $\mathrm{Var}_{\mathbf{epi}}$ (across masks) from $\mathrm{Var}_{\mathbf{alea}}$ (across ICs)



$$\mathrm{Var}_{\mathrm{epi}}(t) = \mathbb{E}_{z_0^i}\big[\mathrm{Var}_m\, y_{m,i}(t)\big]$$
variance across masks at fixed IC

$$\mathrm{Var}_{\mathrm{alea}}(t) = \mathrm{Var}_{z_0^i}\big[\mathbb{E}_m\, y_{m,i}(t)\big]$$
variance across ICs after mask averaging

$$\mathrm{Var}_{\mathrm{tot}}(t) = \mathrm{Var}_{\mathrm{epi}}(t) + \mathrm{Var}_{\mathrm{alea}}(t)$$
law of total variance [Eq. (4)]